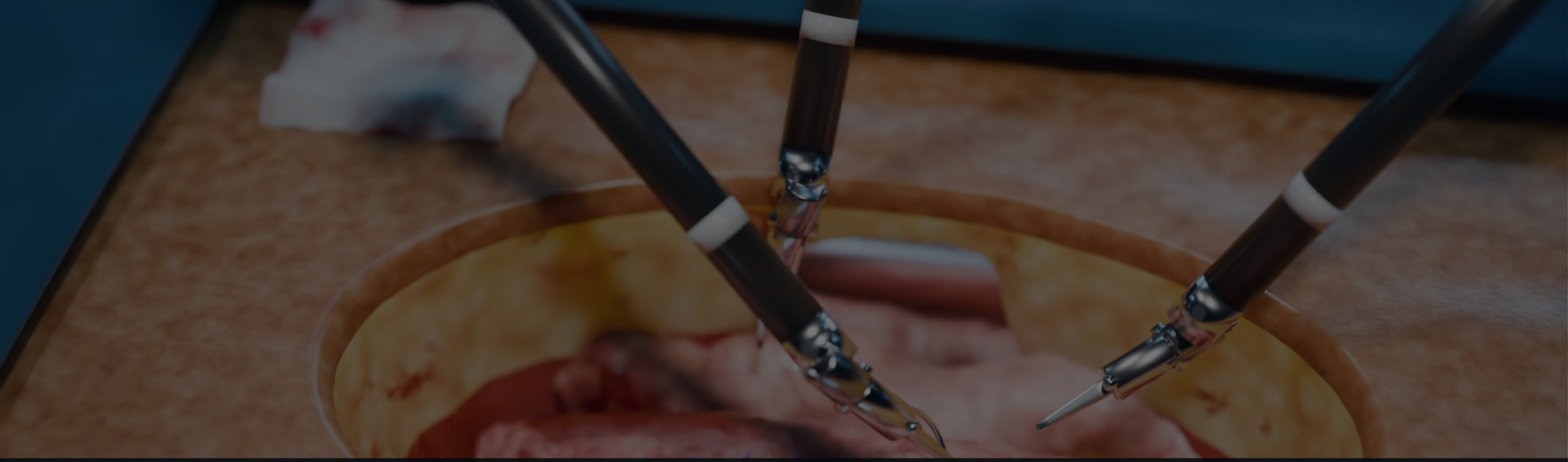


LUMENARA SURGICAL · COMPANY PROFILE 2026

Precision beyond the human hand

Lumenara Arc robotic surgical system · Boston · Munich · Seoul

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A Appendix

One robotic platform for tremor-free, sub-millimetre control in open, laparoscopic and micro-surgery

Physiological tremor, fatigue and the limited range of the human wrist set a ceiling on surgical precision. Lumenara Arc removes it: a real-time motion engine filters tremor and scales motion down by up to 10:1, while cable-driven MicroWrist instruments articulate beyond any human wrist.

0.1 mm
Instrument tip motion resolution

7 DOF
Degrees of freedom at the MicroWrist

10:1
Adjustable motion scaling

4K 3D
Stereoscopic, fluorescence-ready vision

What we do, who we serve, why we are different and how we grow

What we do

Design, manufacture and support the Lumenara Arc robotic surgical system and its MicroWrist instruments.

Who we serve

Hospitals and surgical teams in general, colorectal, urologic, gynecologic, thoracic and reconstructive surgery.

Why we are different

One platform for open, laparoscopic and micro-surgery; up to 10:1 motion scaling; haptic feedback; on-device AI.

How we grow

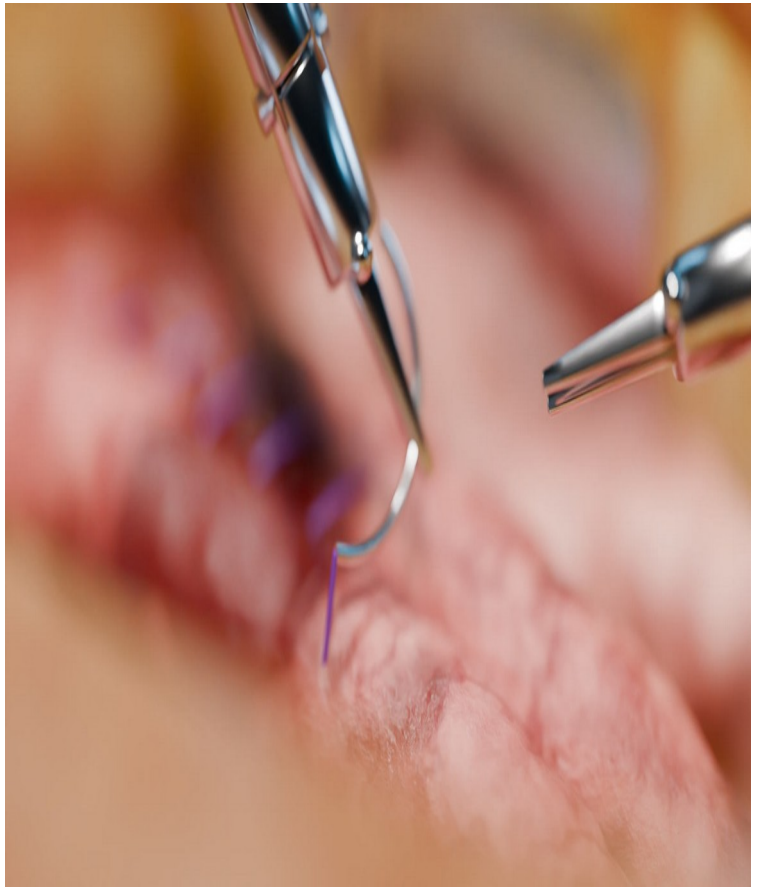
System placements with recurring instrument, service and training revenue, backed by clinical partnerships.

Technology should extend a surgeon's skill, never replace their judgement

When we founded Lumenara, we asked surgeons one question: what would you do if your hands were perfectly steady? Their answers shaped everything we build: sutures placed exactly where intended, vessels repaired below a millimetre, and long reconstructions finished with the precision of the first stitch.

We measure success by what happens after surgery: shorter recoveries, fewer complications and surgeons who can operate at their best for their whole careers.

[Name of CEO]
Co-founder and Chief Executive Officer



Our mission is to give every surgeon the steadiest hands in medicine



Mission

Give every surgeon the steadiest hands in medicine.



Vision

A world where the precision of surgery is limited by judgement, not by anatomy or fatigue.



Values

Patients first · Built with surgeons · Evidence over claims · Safety by design · One team across three continents

Founded in 2016, more than 900 people across three R&D centres

2016

Founded in Boston

900+

Engineers, clinicians and specialists

3

R&D and training centres

40+

Wristed MicroWrist instruments

Boston, USA

Headquarters. Console, robotics and MicroWrist engineering; motion engine and haptics.

Munich, Germany

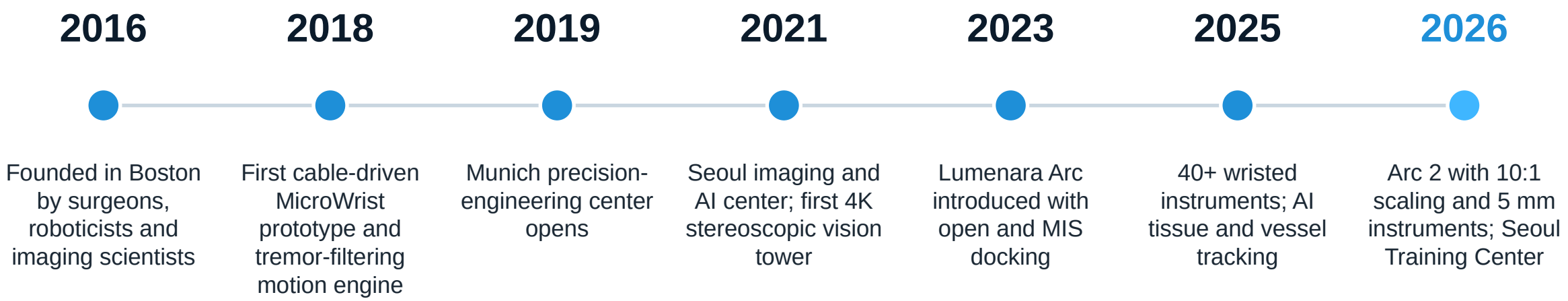
Precision mechanics, sterilization and manufacturing engineering; European clinical affairs.

Seoul, Korea

Imaging, AI scene understanding and software platform; Asia-Pacific training center.

Regulatory status: **[To be provided]**

From a surgeon-led research project to a platform for open, laparoscopic and micro-surgery



Dates before 2026 are illustrative for this concept document.

Surgery now asks for more precision than the unaided hand can reliably deliver

CHALLENGE	WHAT HAPPENS TODAY	LUMENARA ARC
Physiological tremor	8–12 Hz tremor limits fine suturing	**Real-time tremor filtering**
Scale of anatomy	Vessels below 0.8 mm at the edge of manual technique	**10:1 scaling, 0.1 mm tip resolution**
Restricted access	Rigid tools cannot turn corners deep in the body	**7-DOF wrist, ±90° articulation**
Surgeon fatigue	Long cases in awkward postures reduce consistency	**Seated, ergonomic console**
Open vs. MIS divide	Most robots serve port-based surgery only	**Open and MIS docking modes**

Robotic surgery is expanding into more specialties, case types and surgeons per system

Global surgical robotics market

[To be provided from a cited market report]

Annual procedures addressable by Arc

[To be provided]

Priority regions

North America

Installed-base growth, academic partners

Europe

DACH first, from the Munich center

Asia-Pacific

Korea first, from the Seoul Training Center

Market figures are intentionally left blank. Insert data from a cited source before external distribution.

One system for open, laparoscopic and micro-surgical procedures



Arc Console

Where the surgeon operates: seated, focused and in full view of the team.

4K stereoscopic viewer · haptic hand controllers · eye-tracked camera · 1:1–10:1 scaling



Arc Patient Cart

Four slim arms that pivot around a software-defined remote centre of motion.

Open and MIS docking · active collision avoidance · setup in under 5 minutes



Arc Vision Tower

Imaging, energy and intelligence for the whole team.

4K 3D imaging · ICG fluorescence · AI tissue and vessel tracking · case recording

Dock, lock the pivot, operate, verify and learn, in one continuous workflow

1

Dock

Position the cart and dock the arms over the incision or to the ports.

2

Lock the pivot

Each arm locks a remote centre of motion so the body wall takes no strain.

3

Operate

Tremor is filtered and motion scaled in real time from the console.

4

Verify

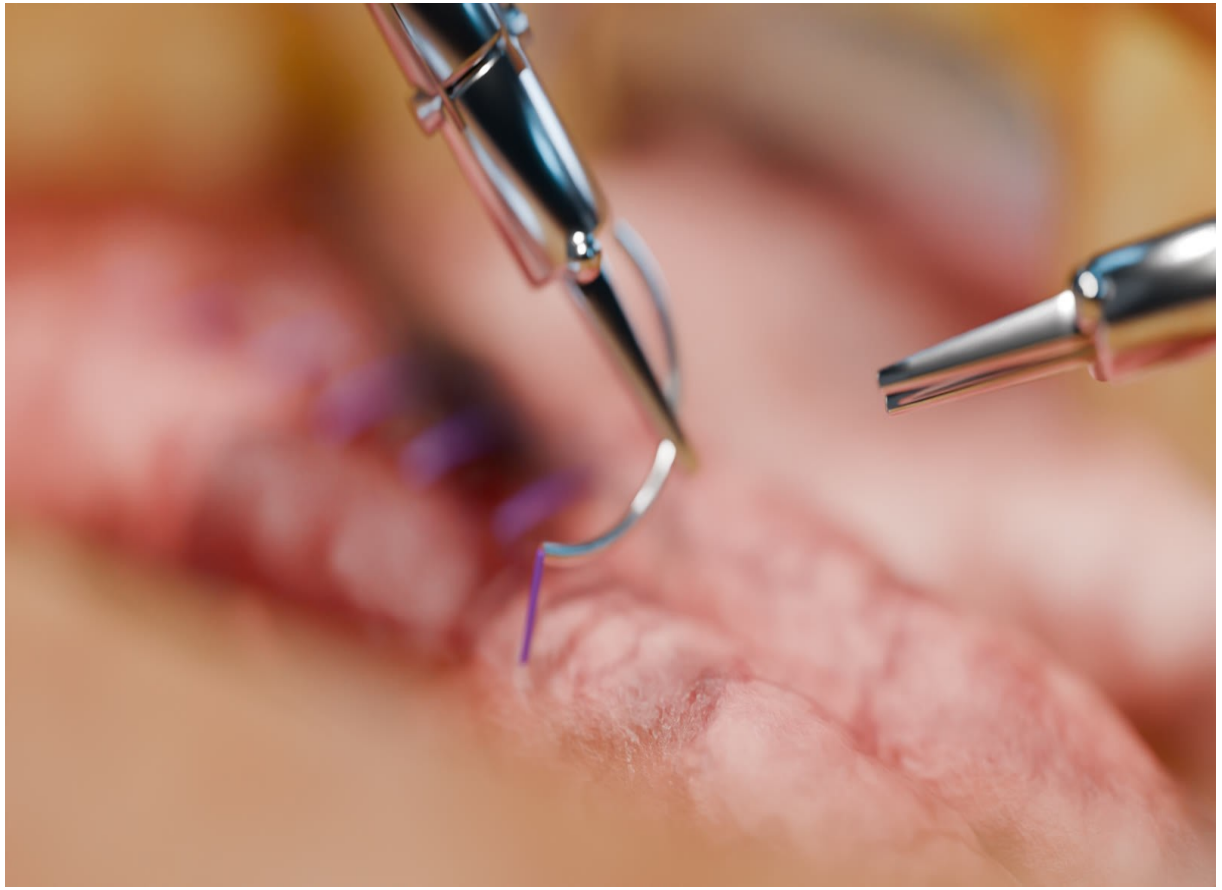
Fluorescence and AI overlays confirm perfusion and critical structures.

5

Learn

The case is recorded securely for review, training and analytics.

A wrist that turns where a human hand can't



Specification	Value
Wrist articulation	±90° pitch · ±90° yaw
Roll	540° continuous
Degrees of freedom	7 at the tip
Shaft diameter	8 mm and 5 mm
Tip resolution	0.1 mm
Instrument library	40+ wristed tools
Usage tracking	Chip-verified life

More than 40 wristed tools across suturing, grasping, dissection and supermicrosurgery

Suturing

Large and micro needle drivers, including suture-cut variants.

Grasping & retraction

Cadiere, fenestrated and atraumatic bowel graspers.

Dissection & energy

Curved and micro scissors, Maryland and fenestrated bipolar, monopolar hook and spatula.

Supermicrosurgery (Arc 2)

5 mm instruments for vessels and lymphatics below 0.8 mm.

Seven technologies that turn a surgeon's intent into steady, precise motion

Motion engine

Filters 8–12 Hz tremor; scales motion 1:1 to 10:1.

Remote centre of motion

Software-locked pivot protects the body wall.

Haptic feedback

Tissue and suture tension returned down to 0.05 N.

AI scene understanding

Real-time segmentation of tissue, vessels and tools.

Fluorescence imaging

Near-infrared ICG reveals perfusion in one click.

Collision-aware arms

Live kinematic model keeps arms and team safe.

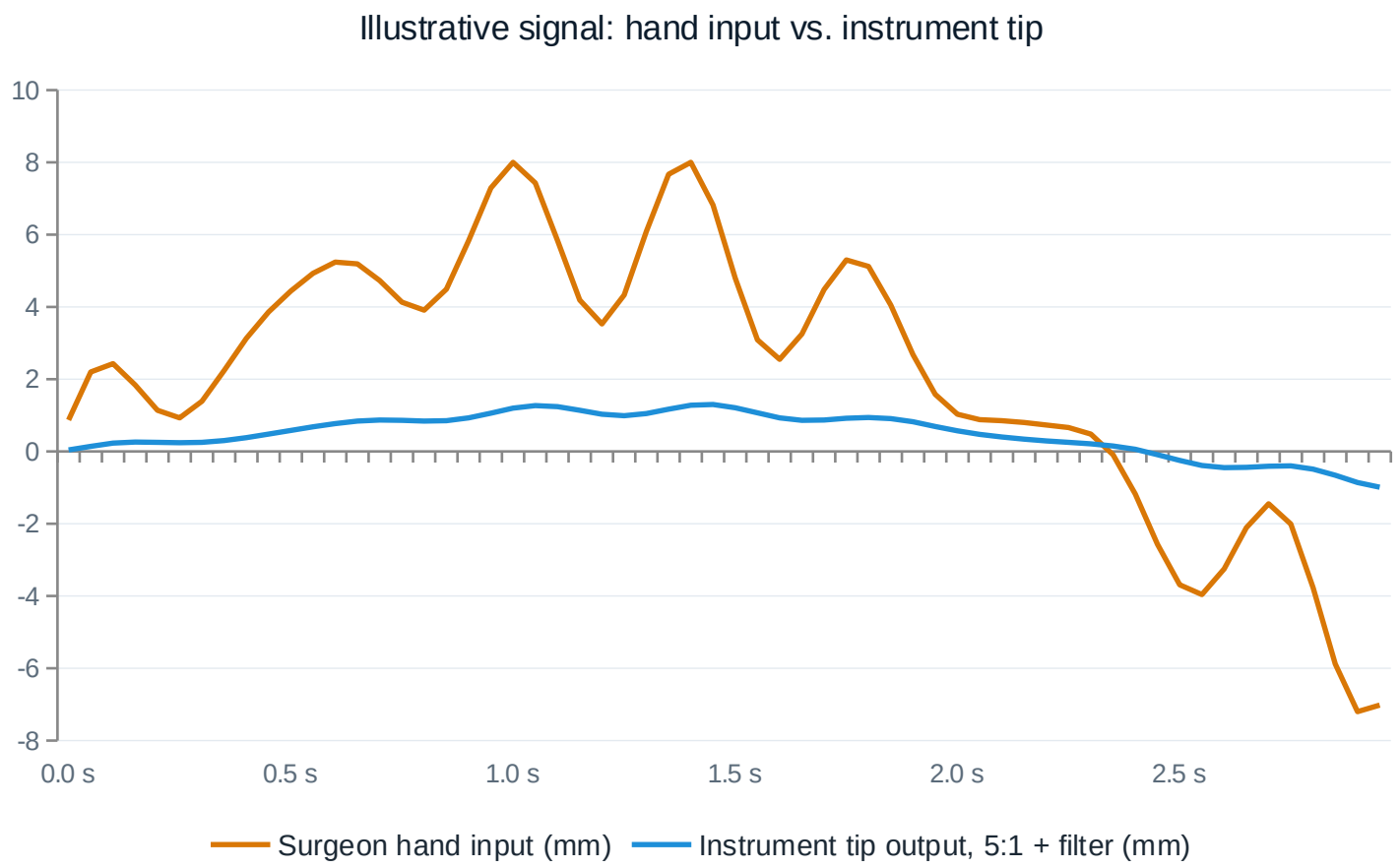
Connected insights

Case analytics for surgeons and teams.

Try it

An interactive motion-engine demo runs on the Technology page of our website.

Tremor is filtered out and motion scaled down before it reaches the instrument tip



1

Capture

Sample the surgeon's hand position at high frequency.

2

Filter

Remove the 8–12 Hz tremor band, keep deliberate motion.

3

Scale

Scale motion down, e.g. 5:1 for suturing, 10:1 for micro.

4

Drive

Compute arm and wrist joints with the RCM locked.

5

Feel

Return tip forces to the hands as haptic feedback.

Built for the procedures that demand the most precision

Specialty	Where Arc adds value	Representative procedures
General & Colorectal	Stable traction and wristed suturing for intracorporeal anastomoses and deep pelvic dissection	Colectomy · LAR · hernia · cholecystectomy
Urology	Nerve-sparing dissection and watertight anastomosis in the confined pelvis	Prostatectomy · partial nephrectomy · pyeloplasty
Gynecology	Visualization and articulation for complex benign and oncologic cases	Hysterectomy · myomectomy · sacrocolpopexy
Thoracic	Articulated access around the hilum through small incisions	Lobectomy · segmentectomy · thymectomy
Micro & Reconstructive	Up to 10:1 scaling for vessels and lymphatics below 0.8 mm	Lymphovenous anastomosis · free flap · nerve repair

Precision only matters when it shows up in recovery



Patients

Smaller incisions, more precise repairs and fewer conversions to open surgery.

WE MEASURE

Length of stay · complications · return to activity



Surgeons

Seated ergonomics, tremor-free control and haptic feedback reduce fatigue.

WE MEASURE

Ergonomic load · fatigue · learning curve



Hospitals

One platform for more specialties and case types, with data to manage the program.

WE MEASURE

Conversion rate · OR time · utilization

Outcome indicators we track with clinical partners

1.2 days

Shorter average hospital stay*

34 %

Fewer conversions to open surgery*

28 %

Lower surgeon-reported fatigue*

96 %

System uptime across installed base*

** Illustrative figures for this concept document. Not clinical data. Replace with peer-reviewed or registry results before external use.*

The only platform designed from day one for open, laparoscopic and micro-surgery

Capability	Typical port-based systems	Lumenara Arc
Surgical approach	Laparoscopic (ports) only	**Open, laparoscopic and micro-surgical**
Motion scaling	Fixed or limited range	**Adjustable 1:1 to 10:1**
Haptic feedback	Limited or none	**Force feedback to 0.05 N**
Supermicrosurgery	Not supported	**5 mm instruments, structures below 0.8 mm**
Intelligence	Imaging only	**On-device AI tissue and vessel tracking**
Setup	Dedicated room, long docking	**Docking in under 5 minutes**

Comparison describes general market positioning and does not refer to any specific competitor product.

Mechanics, software and an ecosystem built with surgeons since 2016

MicroWrist mechanics

Cable-driven wrist design and manufacturing know-how developed since 2018.

Motion engine

Tremor filtering and motion scaling tuned with surgeons across five specialties.

Ecosystem

Training centers, a proctor network and case analytics build long-term relationships.

Intellectual property

Patents filed and granted: **[To be provided]**

A five-step path from first assessment to an optimized robotic program

01

Assessment

Case mix, specialty priorities and program goals with hospital leadership.

02

Program plan

Business case, room planning and a training pathway for the whole team.

03

Installation

Delivery, room integration and validation by Lumenara engineers.

04

Go-live

First cases with on-site clinical specialists and proctors.

05

Optimization

Utilization reviews, new specialties and continuous education.

Service levels: preventive maintenance, remote monitoring, a dedicated clinical specialist during ramp-up; response times **[To be provided]**

From first case to fellowship-level mastery

Simulate

Physics-based VR simulator, 60+ validated exercises, objective scoring.

Practice

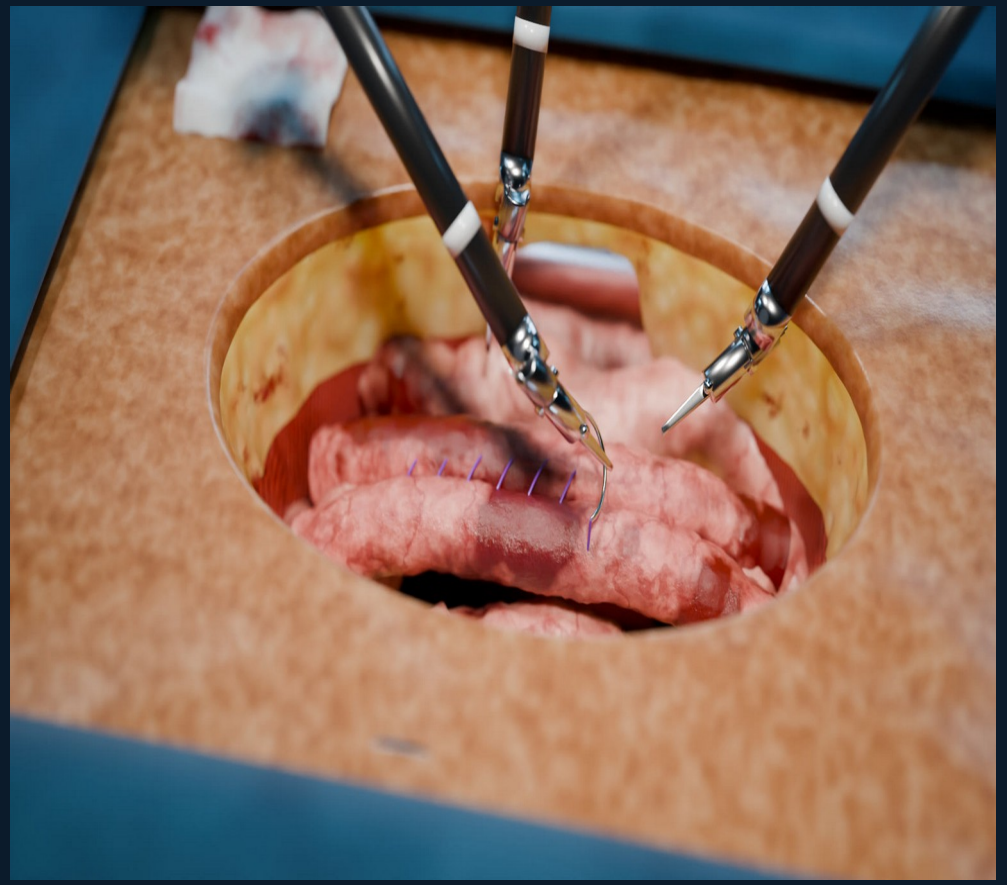
Dry and wet labs at training centers in Boston, Munich and Seoul.

Proctor

Expert proctors join first cases, in person or by tele-mentoring.

Improve

Dashboards benchmark efficiency, economy of motion and outcomes.



A system placement that grows into recurring instrument, service and training revenue

Systems

Console, cart and vision tower

Purchase or lease

Instruments

MicroWrist tools, drapes, accessories

Recurring, per procedure

Service

Maintenance, updates, monitoring

Recurring, annual

Training & data

Courses, proctoring, analytics

Recurring

Acquisition options for hospitals

Purchase

Capital purchase with multi-year service

Lease

Operating lease over the program's life

Usage-based

Per-procedure price incl. instruments and service

Revenue mix and unit economics: **[To be provided]**

From open and MIS surgery today to supermicrosurgery and connected intelligence

Short term

2026–2027

PRODUCT

Arc 2 with 10:1 scaling · 5 mm MicroWrist · expanded AI tracking

MARKET

Seoul Training Center · Asia-Pacific placements

Mid term

2028–2029

PRODUCT

Supermicrosurgery instrument set · procedure-specific AI guidance · tele-mentoring at scale

MARKET

More European and Asia-Pacific markets · academic partnerships

Long term

2030+

PRODUCT

Assisted autonomy for well-defined subtasks, always under surgeon control

MARKET

Global installed base · outcome-based service models

Roadmap items are forward-looking and subject to regulatory clearance in each market.

See Lumenara Arc in your operating room



Demonstration

Hands-on session with your surgical team, at your hospital or a Lumenara Training Center.



Clinical partnership

Joint studies, registries and fellowship programs.



Distribution

Regional sales and service partnerships.

WEBSITE

robot-surgery-ai.vercel.app

REQUEST A DEMO

robot-surgery-ai.vercel.app/demo/

EMAIL

[To be provided]

PHONE

[To be provided]

OFFICES

Boston (HQ) · Munich · Seoul

Lumenara Arc system specifications

Specification	Value
Arms	4, collision-aware, remote centre of motion
Docking modes	Open (over incision) and MIS (ports)
Motion scaling	1:1 to 10:1, adjustable
Tremor filtering	8–12 Hz band, 4 levels (Off / Low / High / Max)
Tip motion resolution	0.1 mm
Haptic feedback	To 0.05 N
Vision	4K stereoscopic 3D, near-infrared ICG fluorescence
Instruments	40+ MicroWrist tools, 8 mm and 5 mm
Setup time	Under 5 minutes

Key terms used in this profile

Term	Meaning
Anastomosis	Surgical connection of two tubular structures, such as bowel ends or vessels
DOF	Degrees of freedom: independent ways an instrument can move
ICG fluorescence	Near-infrared imaging with indocyanine green to show blood flow
Laparotomy	An open incision into the abdominal cavity
MIS	Minimally invasive surgery through small ports
Motion scaling	Reducing hand motion to a smaller instrument motion, e.g. 5:1
RCM	Remote centre of motion: fixed instrument pivot at the incision or port

Notice: concept company profile. Lumenara Surgical, Lumenara Arc and MicroWrist are fictional names. Specifications, outcome indicators and milestones are illustrative and do not describe an actual medical device or clinical results. Fields marked [To be provided] must be completed before external use. Imagery was procedurally modelled and rendered in Blender.